The Smart Drone Delivery System

1. The Custom Exception (InvalidCargoException) For the first requirement, I created a custom checked exception by extending the base Exception class. Basically, I set up a constructor that takes a string message so that whenever an invalid cargo slot is requested, it can pass a clear error description up the chain.

2. Array Validation (inspectCargoSlot) I initialized a static array called cargoWeights with four preset values. The inspectCargoSlot method takes the slot index you type in and checks if it's less than 0 or bigger than the array's length. If it is out of bounds, it throws my InvalidCargoException. If it's valid however, it just prints out the weight at THAT specific slot.

3. Flight Efficiency Calculation (calculateEfficiency) For this part, this method divides the total energy used by the distance traveled. I didn't put a try-catch block inside this method; instead, I explicitly checked if the distance is 0.0. If it is, I throw an ArithmeticException so it bubbles straight up to the main menu loop instead of handling it locally.

4. The Main Console Loop (main) Everything runs inside a while loop using a Scanner so the menu keeps popping up until you choose to exit. I used a switch statement to handle the user's menu choices. I also wrapped the calls to inspectCargoSlot and calculateEfficiency in try-catch blocks. That way, if a custom exception or an arithmetic exception gets thrown, the program catches it, prints the error message, and goes right back to the menu.

Test Cases:

```
Problems  Output  Debug Console  Terminal  Ports

PS C:\OOP_Projects> ^C
PS C:\OOP_Projects>
PS C:\OOP_Projects>  c:; cd 'c:\OOP_Projects'; & 'C:\Program Files\Java\jdk-17\bin\jav
0178fba74a52d4c02a46b6\redhat.java\jdt_ws\OOP_Projects_f1cbe841\bin' 'InteractiveDrone

=== Smart Drone Control Console ===
1. Inspect Cargo Slot
2. Calculate Flight Efficiency
3. Exit
Choose an option: 3
Shutting down drone control console. Goodbye.
PS C:\OOP_Projects>
```

```
Problems  Output  Debug Console  Terminal  Ports

PS C:\OOP_Projects>  & 'C:\Program Files\Java\jdk-17\bin\java.exe' '-XX:+ShowCo
hat.java\jdt_ws\OOP_Projects_f1cbe841\bin' 'InteractiveDroneSystem'

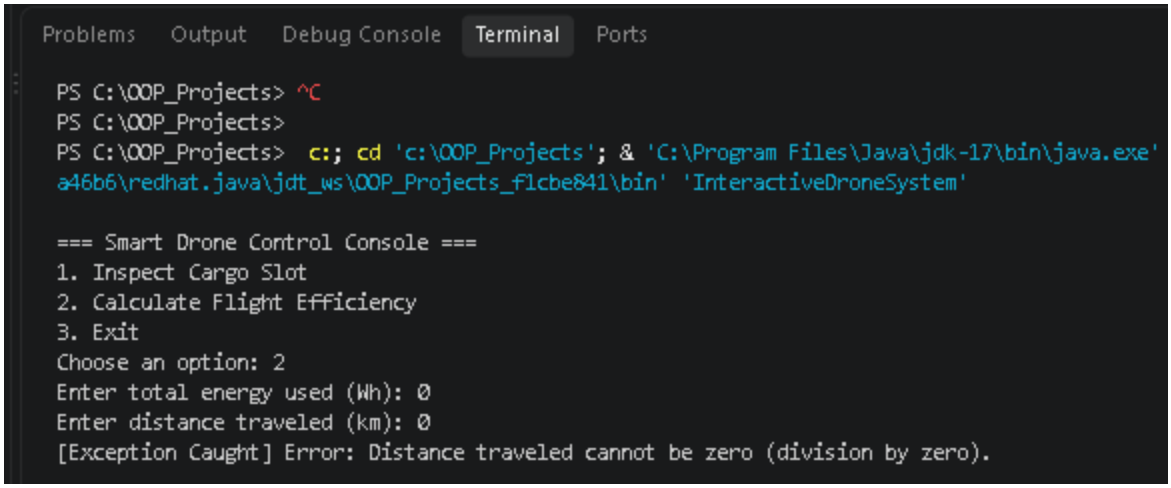
=== Smart Drone Control Console ===
1. Inspect Cargo Slot
2. Calculate Flight Efficiency
3. Exit
Choose an option: hi im jian
Invalid input. Please enter a valid menu number.
```

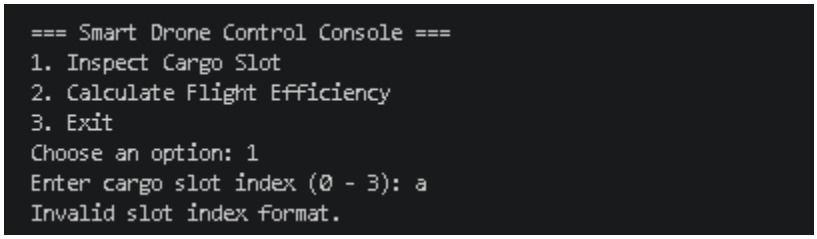
```
Problems  Output  Debug Console  Terminal  Ports

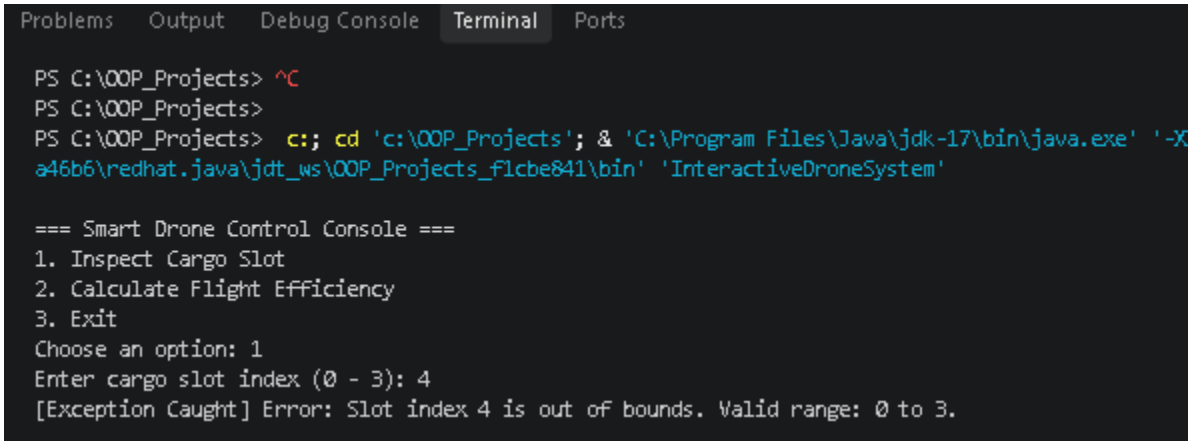
PS C:\OOP_Projects> ^C
PS C:\OOP_Projects>
PS C:\OOP_Projects>  c:; cd 'c:\OOP_Projects'; & 'C:\Program Files\Java\jdk-17\bin\jav
a46b6\redhat.java\jdt_ws\OOP_Projects_f1cbe841\bin' 'InteractiveDroneSystem'

=== Smart Drone Control Console ===
1. Inspect Cargo Slot
2. Calculate Flight Efficiency
3. Exit
Choose an option: 2
Enter total energy used (Wh): 150
Enter distance traveled (km): 0
[Exception Caught] Error: Distance traveled cannot be zero (division by zero).
```

```
=== Smart Drone Control Console ===
1. Inspect Cargo Slot
2. Calculate Flight Efficiency
3. Exit
Choose an option: 2
Enter total energy used (Wh): a
Invalid numeric input. Enter energy used (Wh):
a
```


Problems
Output
Debug Console
Terminal
Ports
PS C:\OOP_Projects> ^C
PS C:\OOP_Projects>
PS C:\OOP_Projects> c:; cd 'c:\OOP_Projects'; & 'C:\Program Files\Java\jdk-17\bin\java.exe'
a46b6\redhat.java\jdt_ws\OOP_Projects_f1cbe841\bin' 'InteractiveDroneSystem'
=== Smart Drone Control Console ===
1. Inspect Cargo Slot
2. Calculate Flight Efficiency
3. Exit
Choose an option: 2
Enter total energy used (Wh): 0
Enter distance traveled (km): 0
[Exception Caught] Error: Distance traveled cannot be zero (division by zero).


=== Smart Drone Control Console ===
1. Inspect Cargo Slot
2. Calculate Flight Efficiency
3. Exit
Choose an option: 1
Enter cargo slot index (0 - 3): a
Invalid slot index format.


Problems
Output
Debug Console
Terminal
Ports
PS C:\OOP_Projects> ^C
PS C:\OOP_Projects>
PS C:\OOP_Projects> c:; cd 'c:\OOP_Projects'; & 'C:\Program Files\Java\jdk-17\bin\java.exe' '-X
a46b6\redhat.java\jdt_ws\OOP_Projects_f1cbe841\bin' 'InteractiveDroneSystem'
=== Smart Drone Control Console ===
1. Inspect Cargo Slot
2. Calculate Flight Efficiency
3. Exit
Choose an option: 1
Enter cargo slot index (0 - 3): 4
[Exception Caught] Error: Slot index 4 is out of bounds. Valid range: 0 to 3.

```
=== Smart Drone Control Console ===
1. Inspect Cargo Slot
2. Calculate Flight Efficiency
3. Exit
Choose an option: 1
Enter cargo slot index (0 - 3): 3
Cargo weight at slot 3: 10.4 kg
```

```
=== Smart Drone Control Console ===
1. Inspect Cargo Slot
2. Calculate Flight Efficiency
3. Exit
Choose an option: 1
Enter cargo slot index (0 - 3): 2
Cargo weight at slot 2: 15.2 kg
```

Problems Output Debug Console **Terminal** Ports

```
PS C:\OOP_Projects> ^C
PS C:\OOP_Projects>
PS C:\OOP_Projects>  c:; cd 'c:\OOP_Projects'; & 'C:\Program Files\Java\jdk-17\bin\java.exe'
a46b6\redhat.java\jdt_ws\OOP_Projects_f1cbe841\bin' 'InteractiveDroneSystem'

=== Smart Drone Control Console ===
1. Inspect Cargo Slot
2. Calculate Flight Efficiency
3. Exit
Choose an option: 1
Enter cargo slot index (0 - 3): 0
Cargo weight at slot 0: 12.5 kg

=== Smart Drone Control Console ===
1. Inspect Cargo Slot
2. Calculate Flight Efficiency
3. Exit
Choose an option:
```